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M168_Notes_7.1

CHAPTER 7: PROBABILITY

7.1: Experiments, Sample Spaces and Events

An ***experiment*** is an activity with an observable result. Tossing coins, rolling dice and choosing cards are all probability experiments.

The result of the experiment is called the ***outcome*** or ***sample point***. The set of all outcomes or sample points is called the ***sample space*** of the experiment.

Example
What is the sample space for flipping a fair coin? Rolling a 6-sided die?

$S = \{H, T\}$, $S = \{1, \ldots, 6\}$

An ***event*** is a subset of a sample space. That is, an event can contain one or more outcomes that are in the sample space.

Example
What are all possible events for the experiment of flipping a fair coin?

$\phi, \{H\}, \{T\}, \{H, T\} \rightarrow 4 = 2^2$

Example
How many events are possible when a six-sided die is rolled?

2^6

Example

What is the sample space for flipping a fair coin three times?

S={HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}

Find the event E where $E = \{x|x$ has exactly one head$\}$

{HTT, THT, TTH}

Find the event E where $E = \{x|x$ has two or more heads$\}$

{HHH, HHT, HTH, THH}

Find the event E where $E = \{x|x$ has more than 3 heads$\}$

{ }, ϕ

A sample space in which each of the outcomes has the same chance of occurring is called a UNIFORM SAMPLE SPACE.

Example

A bowl contains the letters AGGIES. How many outcomes are in the uniform sample space?

{A, G, G, I, E, S}, $n(S)=6$

What is the uniform sample space for rolling two fair six-sided dice?

1~1	2~1	3~1	4~1	5~1	6~1
1~2	2~2	3~2	4~2	5~2	6~2
1~3	2~3	3~3	4~3	5~3	6~3
1~4	2~4	3~4	4~4	5~4	6~4
1~5	2~5	3~5	4~5	5~5	6~5
1~6	2~6	3~6	4~6	5~6	6~6

$n(S) = 36$

$4 \sim 3 \rightarrow$ 4 as outcome on the first roll.
3 " " " " second roll.

These sample spaces have all been finite. That is, we can list all the elements. An infinite sample space has to be described; you can't list all the elements:

Example
What is the sample space for the time spent working on a homework set?

$S = \{ t \mid t \geq 0,\ t \text{ in minutes} \}$

Describe the event of spending between one and two hours on a homework set.

$E = \{ t \mid 60 < t < 120 \}$